

MedGraphRAG: A Safety-First Hybrid Graph Retrieval-Augmented Generation System for Offline Clinical Decision Support

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Abstract

Background and Objective. Large language models (LLMs) deployed in clinical decision support face a safety-utility trade-off: unconstrained answering yields unacceptable rates of wrong assertions, whereas excessive refusal renders systems inert. Existing retrieval-augmented generation (RAG) frameworks lack mechanistic, bounded safety controls deployable on institutional hardware. We present MedGraphRAG, an offline hybrid GraphRAG system that reframes clinical AI evaluation from unconstrained accuracy maximisation to bounded wrong-assertion rate (WAR).

Methods. MedGraphRAG couples dense vector retrieval (FAISS IVF-PQ with MedCPT embeddings) and lexical retrieval (BM25) with an embedded property graph (Kuzu) enriched with 632,931 relational edges (617,219 NEGATES and 15,712 TEMPORAL_BEFORE). Dual candi-

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date streams are merged via Reciprocal Rank Fusion and re-ranked with an additive relation-aware score ($\gamma = 0.15$). A beta-gated verification layer enforces a strict $\text{WAR} \leq 10\%$ constraint by combining evidence faithfulness (φ) and graph consistency (S_g): $V = \beta\varphi + (1 - \beta)S_g$. The system executes entirely offline on a commodity laptop (Ryzen 7 7840HS / RTX 3050 6 GB) using Qwen2.5-7B-Instruct Q4_K_M. Private health data are encrypted with AES-256-GCM and HKDF-derived keys. Retrospective evaluation was performed on MedQA-US ($N = 500$, seed 42, $T = 0.0$).

Results. At the calibrated safe threshold ($\tau^* = 0.80$), Graph-Only mode (M2) achieves 60.8% precision-on-answered with $\text{WAR} = 8.0\%$ (79.6% refusal). Disabling the gate drives WAR to 44.6% across all 500 queries. Deterministic query rewriting expands answered coverage by 3.2–3.4 pp while maintaining bounded error. Hybrid retrieval completes in 455.7 ms (95% CI: [388.4, 495.8] ms); full end-to-end pipeline latency is 4,083 ms (M2) to 12,667 ms (M4).

Conclusion. MedGraphRAG demonstrates that bounded clinical error and interpretable offline decision support are jointly achievable on consumer hardware without cloud dependencies or GPU clusters.

Keywords: GraphRAG, clinical-NLP, hallucination-mitigation, knowledge-graph, retrieval-augmented-generation, decision-support, safety

1. Introduction

Deploying large language models (LLMs) as point-of-care clinical assistants offers direct utility for diagnostic and therapeutic workflows (Singhal et al., 2023; Thirunavukarasu et al., 2023). However, the tendency of ungrounded parametric architectures to hallucinate plausible yet factually incorrect assertions poses direct patient-safety hazards (Bai et al., 2022). In high-stakes medicine, confident assertions regarding contraindicated medications, misidentified diagnostic criteria, or inverted pathological chronologies can cause serious clinical harm.

To mitigate hallucinations, retrieval-augmented generation (RAG) grounds language generation on passages retrieved from external corpora (Lewis et al., 2020; Gu et al., 2021). Standard medical RAG frameworks rely on dense bi-encoders (Jin et al., 2023; Gao et al., 2023), which match semantic similarity effectively but struggle with complex clinical syntax. Specifically, vector embeddings frequently compress or lose negation scope (e.g., conflating “no evidence of pulmonary embolism” with “pulmonary embolism”) and fail to enforce multi-step temporal chronologies required in differential diagnosis (Han et al., 2022; Mallen et al., 2023).

Knowledge graphs provide a complementary symbolic representation, structuring clinical entities and ontological dependencies into explicit triples (Bodenreider, 2004; Yasunaga et al., 2021). Recent hybrid GraphRAG systems combine graph structures with textual retrieval (Edge et al., 2024; He et al., 2024; Wu et al., 2024) (Wu et al. share the MedGraphRAG name

24 but focus on entity-centric graph construction; our architecture emphasises
25 hybrid retrieval fusion and dual-gated verification). Nevertheless, existing
26 medical GraphRAG implementations exhibit three fundamental limitations:

27 First, current evaluations focus almost exclusively on unconstrained ac-
28 curacy (e.g., aggregate multiple-choice scores), omitting explicit bounds on
29 incorrect statements. In clinical practice, an assistant that answers 60% of
30 questions correctly while making wrong assertions on 40% is clinically un-
31 acceptable. Safe deployment requires a constrained-optimisation framework
32 that caps the wrong-assertion rate (WAR) below an acceptable tolerance
33 ϵ , treating refusal (abstention) as an active safety mechanism rather than
34 system failure.

35 Second, existing biomedical knowledge graphs capture predominantly pos-
36 itive ontological relations (e.g., TREATS or CAUSES), neglecting negation
37 and temporal progression. Without these primitives, retrieval graphs cannot
38 detect when retrieved passages contain superseded diagnoses or explicit ex-
39 clusions.

40 Third, most GraphRAG systems rely on cloud-hosted language models
41 (e.g., GPT-4) and server clusters. This architecture violates patient data pri-
42 vacy regulations (e.g., HIPAA, GDPR) that prohibit transmitting protected
43 health information (PHI) beyond institutional boundaries. Practical point-
44 of-care deployment requires secure, sub-second hybrid retrieval and inference
45 executing locally on commodity hardware.

46 We present **MedGraphRAG**, an offline hybrid GraphRAG system de-

47 signed for point-of-care clinical decision support. Our primary contributions
48 are:

- 49 1. **Safety-Constrained Pareto Calibration:** A verification framework
50 optimising answered accuracy subject to $\text{WAR} \leq 10\%$. Across 15
51 threshold settings on MedQA-US ($N = 500$), we establish the Pareto
52 frontier, demonstrating that ungated generation yields unacceptable
53 error rates ($\approx 44\%$).
- 54 2. **Relational Graph Enrichment:** Ingestion of 632,931 negation and
55 temporal relational edges into an embedded Kuzu graph, demonstrat-
56 ing mechanistically how symbolic consistency checks overturn halluci-
57 nations bypassing dense retrieval.
- 58 3. **Deterministic Query Rewriting and**
59 **Relation Reranking:** A temperature-zero query normaliser improv-
60 ing answered coverage by 3.2–3.4 pp at matched thresholds, combined
61 with a relation-aware reranker boosting Graph-Hit@5 by 13.6 pp.
- 62 4. **Hardened Offline Stack on Consumer Hardware:** A self-
63 contained system operating on an NVIDIA RTX 3050 (6 GB) laptop
64 GPU, achieving hybrid retrieval in 455.7 ms with authenticated
65 AES-256-GCM encryption and a resolved 13-point security audit.

66 Supplementary materials provide the DECIDE-AI applicability mapping,
67 IJMI ML checklist, and extended tables; configuration details appear in Ap-
68 pendix A.

69 **2. Methods**

70 *2.1. System Architecture*

71 Figure 1 shows the MedGraphRAG inference pipeline. Stage 1 determin-
72 istically standardises input queries into clinical phrasing. Stage 2 executes
73 dual-stream dense and lexical retrieval, combining candidates via Reciprocal
74 Rank Fusion and relation-aware graph reranking. Stage 3 expands clinical
75 entities within the embedded Kuzu graph. Stage 4 computes composite beta-
76 gated verification scores, triggering structured refusals for low-confidence
77 queries. Stage 5 generates evidence-grounded responses using a quantised
78 LLM isolated behind a GPU mutex lock.

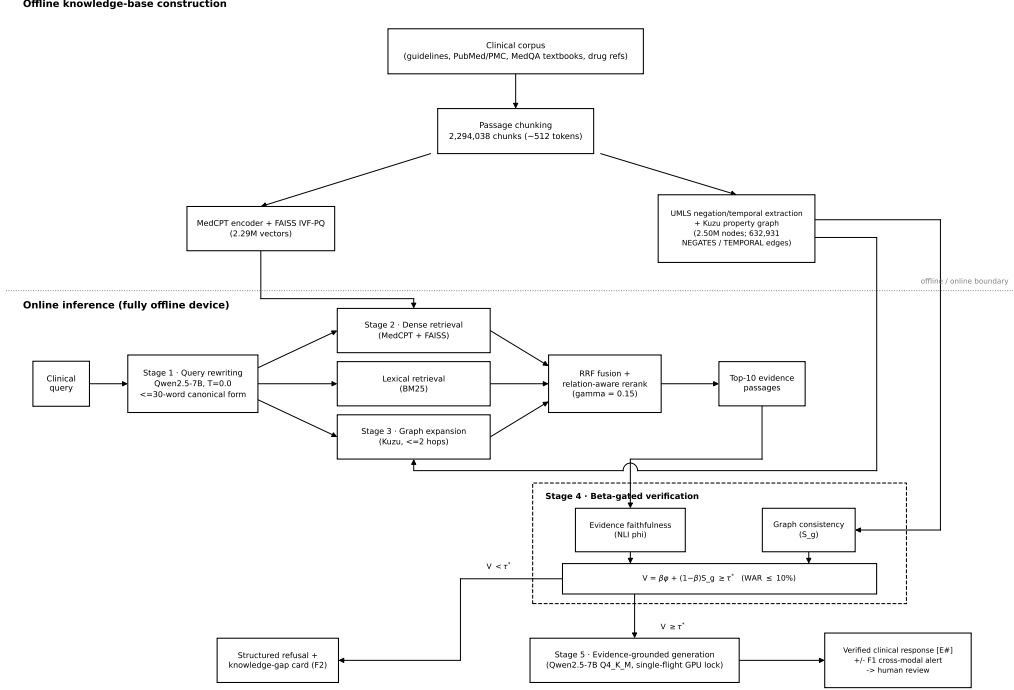


Figure 1: MedGraphRAG five-stage offline pipeline architecture. Stage 1 executes deterministic query rewriting ($T = 0.0, \leq 30$ words). Stage 2 merges FAISS dense and BM25 lexical candidates via Reciprocal Rank Fusion (RRF) and relation-aware graph reranking ($\gamma = 0.15$). Stage 3 traverses the Kuzu property graph. Stage 4 applies beta-gated verification ($\beta = 0.7$) enforcing $\text{WAR} \leq 10\%$. Stage 5 serialises LLM generation behind a hardware mutex lock.

79 2.2. Deterministic Query Rewriting

80 Patient case vignettes and clinical inquiries often carry conversational
81 phrasing, narrative patient histories, and extraneous details that impede
82 biomedical dense retrieval. Dense bi-encoders trained on scientific literature
83 struggle with multi-sentence case presentations where incidental biographical
84 history obscures the diagnostic dilemma. Lexical search methods also face
85 vocabulary mismatch when colloquial symptom descriptions diverge from for-

mal medical nomenclature.

To standardise heterogeneous queries without stochastic variance, Stage 1 uses Qwen2.5-7B-Instruct executing strictly at temperature $T = 0.0$. The rewriter uses a structured system prompt directing the model to extract the primary pathological condition, salient diagnostic findings, and therapeutic targets into a canonical clinical summary capped at 30 words. For example, a narrative vignette describing an elderly patient who experiences severe rotational vertigo triggered by lateral head motion is transformed into the canonical clinical query: “benign paroxysmal positional vertigo diagnostic criteria and Dix-Hallpike confirmatory maneuvers.” Setting temperature $T = 0.0$ alongside greedy decoding guarantees mathematical determinism: identical clinical inputs generate bit-identical textual reformulations across successive executions, fulfilling mandatory auditability and reproducibility requirements for clinical decision support software.

2.3. Dual-Stream Hybrid Retrieval and Reranking

Retrieval operates over an extensive clinical reference corpus partitioned into coherent passages of approximately 512 tokens. The dense retrieval stream encodes passages using the MedCPT query and article bi-encoder architecture (Jin et al., 2023), which was pre-trained on hundreds of millions of PubMed search query-article click logs to optimise zero-shot biomedical retrieval. Chunk vectors are indexed and searched using FAISS IVF-PQ (Inverted File with Product Quantisation; Johnson et al., 2021), partitioning

the vector space into coarse Voronoi centroids to achieve sub-linear retrieval speed. Concurrently, a parallel lexical stream evaluates candidate documents using the classical BM25 probabilistic relevance framework (Robertson and Zaragoza, 2009), operating over a dense candidate pool of 200 retrieved documents.

Because dense and lexical retrievers exhibit complementary error profiles (dense models capture latent semantic affinity, whereas lexical models enforce exact keyword and numerical matches), their respective rank distributions R_{dense} and R_{lex} are integrated via Reciprocal Rank Fusion (RRF; Cormack et al., 2009):

$$s_{\text{fused}}(d) = \sum_{m \in \{\text{dense}, \text{lex}\}} \frac{1}{k_{\text{rrf}} + r_m(d)}, \quad (1)$$

where $r_m(d)$ denotes the ordinal rank of candidate passage d within stream m , and $k_{\text{rrf}} = 60$ represents the standard monotonic smoothing constant that prevents top-ranked outliers in a single stream from dominating the fused ranking.

To bias retrieval toward passages with corroborated topological connections to the clinical entities identified in the query, we implement an additive relation-aware graph reranking function:

$$s_{\text{hybrid}}(d) = s_{\text{fused}}(d) + \gamma \cdot r_{\text{rel}}(d), \quad (2)$$

where $r_{\text{rel}}(d) = \sum_{e \in \mathcal{E}(d)} w(\text{type}(e))$ aggregates the domain-weighted edge counts connecting entities cited within candidate passage d to the query

seed entities within the property graph. The relative importance of edge categories is parameterised by clinical specificity weights (e.g., pharmacological treatments and adverse effects receive weight 1.0, diagnostic lab associations and negation edges receive 0.8, temporal sequences receive 0.6, and broad taxonomic hierarchies receive 0.3; Appendix A). The global multiplier $\gamma = 0.15$ balances semantic similarity with structural graph reinforcement, calibrated empirically via controlled ablations (Section 3.3).

2.4. Relational Knowledge Graph Enrichment

Structured relations are stored in Kuzu (Feng et al., 2023), an embeddable, disk-backed graph database supporting openCypher queries. While standard medical graphs capture static disease-symptom linkages, clinical decision support requires modelling negation and temporal order.

We processed 2,294,038 text chunks using rule-guided linguistic parsing grounded in UMLS (Bodenreider, 2004). Operating at 0.2329 ms per chunk (534.45 s total wall-clock), the pipeline ingested **632,931** directed relational edges:

- **NEGATES Edges (617,219):** Linking text chunks to entities bearing explicit negation cues (“no evidence of,” “ruled out,” “contraindicated in”), comprising 111,720 Chunk $\rightarrow$ Disease, 21,001 Chunk $\rightarrow$ Drug, and 484,498 Chunk $\rightarrow$ OntologyTerm edges.
- **TEMPORAL_BEFORE Edges (15,712):** Linking clinical concepts via chronological sequence cues (“prior to,” “following,”

149 “within one hour of”), comprising 12,225 Disease $\rightarrow$ Disease and 3,487
150 Drug $\rightarrow$ Drug edges.

151 Manual auditing of sampled endpoints confirmed high directional validity. A
152 cryptographic SHA-256 retrieval signature verified 0.00% retrieval drift on
153 baseline queries post-ingestion (Appendix A).

154 2.5. Beta-Gated Verification and Pareto Threshold Calibration

155 Standard RAG architectures lack explicit confidence gates to arbitrate
156 whether retrieved evidence warrants generating an answer. Unhedged lan-
157 guage models produce fluent assertions even when underlying evidence is
158 tenuous, contradictory, or absent. To prevent clinical hallucinations, Med-
159 GraphRAG adds a dual-evidence verification layer that evaluates candidate
160 response fidelity prior to releasing output.

161 For each candidate assertion, the verification module evaluates a compos-
162 ite verification metric $V \in [0, 1]$ that combines passage-level entailment with
163 symbolic graph consistency:

$$V = \beta \varphi + (1 - \beta) S_g, \quad (3)$$

164 where $\varphi \in [0, 1]$ denotes an NLI-derived faithfulness score measuring the de-
165 gree of semantic entailment between the generated assertion and the retrieved
166 reference passages (operating via normalised MedCPT embedding cosine sim-
167 ilarity as an efficient deterministic proxy when an independent cross-encoder

168 NLI judge is omitted), and $S_g \in [0, 1]$ denotes the structural graph consistency score. The graph consistency metric S_g evaluates the ratio of asserted
 169 clinical entity triples that are corroborated by non-negated multi-hop paths
 170 within a 2-hop radius in Kuzu, while applying an immediate penalising discount
 171 whenever an asserted entity matches a NEGATES edge. The balance
 172 hyperparameter $\beta = 0.70$ allocates primary decision weight to passage evidence
 173 while preserving strong sensitivity to symbolic graph contradictions.

175 When the composite score falls below a calibrated refusal threshold τ ,
 176 generation is suppressed, and the system emits a structured clinical refusal
 177 detailing the precise nature of the missing evidence. We calibrate τ through
 178 a constrained optimisation problem:

$$\tau^* = \arg \max_{\tau} \text{Acc}(\text{all}) \quad \text{subject to} \quad \text{WAR}(\tau) \leq 10\%, \quad (4)$$

179 where $\text{Acc}(\text{all})$ represents the proportion of correct responses across the entire
 180 evaluation cohort ($N = 500$), and WAR represents the proportion of wrong
 181 assertions across the same cohort. We sweep 15 discrete threshold candidate
 182 points ($\tau \in \{0.10, 0.15, 0.20, \dots, 0.80\}$), identifying the Pareto-optimal operating
 183 threshold τ^* that maximises clinically useful answers while strictly
 184 guaranteeing that erroneous assertions remain within the institutional safety
 185 threshold.

186 2.6. Safety Guardrails

187 Two deterministic guardrails operate at sub-millisecond latency:

- 188 • **F1: Cross-Modal Discrepancy Guardrail.** In multimodal con-
189 texts, Rule A detects conflicts when an imaging finding score ≥ 0.60
190 contradicts a text negation cue for the same finding (e.g., visual pneu-
191 mothorax detection of 0.88 versus report text “clear lungs”). Rule B
192 detects positive textual assertions that contradict negated imaging ob-
193 servations. F1 escalates conflicts by issuing a high-severity alert recom-
194 mending specialist review rather than attempting autonomous resolu-
195 tion.
- 196 • **F2: Epistemic Knowledge-Gap Mapper.** When a query is refused,
197 F2 classifies the failure into three gap types matching the implemented
198 taxonomy: G1 (`corpus_retrieval`, top fused retrieval score < 0.02),
199 G2 (`graph_coverage`, Kùzu graph traversal returns a neutral check
200 with no direct relational assertion), or G3 (`evidence_faithfulness`,
201 verification faithfulness score $\varphi < 0.50$). This categorisation is provided
202 in the structured refusal output to guide clinical follow-up.

203 2.7. Offline Cryptographic Security Architecture

204 To ensure institutional compliance without external cloud infrastructure,
205 user-uploaded records are isolated in per-user directories. Payloads are en-
206 crypted at rest with AES-256-GCM using unique 12-byte nonces. Keys are
207 derived dynamically via HKDF-SHA-256 (Krawczyk and Eronen, 2010) com-
208 bining a server secret, user ID, and document ID. File permissions are set
209 to 0700 (directories) and 0600 (files). All Kuzu queries use parameterised

openCypher statements to eliminate Cypher injection. A 13-point security audit resolved all vulnerabilities prior to release (Supplementary Table S7).

2.8. Evaluation Protocol and Metrics

Retrospective evaluation was performed on MedQA-US (Jin et al., 2021), comprising USMLE-derived multiple-choice questions ($N = 500$, seed 42, $T = 0.0$). We evaluated four modes: **M1** Evidence-Only ($\beta = 1.0$), **M2** Graph-Only ($\beta = 0.0$), **M3** Combined Hybrid ($\beta = 0.7$), and **M4** Hybrid Rerank ($\beta = 0.7, \gamma = 0.15$). Metrics include: Acc(all) (fraction correct over N), Acc(ans) (precision on answered questions), WAR (fraction incorrect over N), and Refusal Rate (fraction refused). Non-parametric bootstrap 95% confidence intervals were calculated using 1,000 resamples.

3. Results

3.1. Safe Operating Points and WAR Constraint

Table 1 summarises the optimal operating parameters established across the four evaluation modes under the institutional safety constraint $\text{WAR} \leq 10.0\%$. Graph-Only mode (M2) attains the highest precision-on-answered at its optimal operating threshold $\tau^* = 0.80$, achieving $\text{Acc}(\text{ans}) = 60.8\%$ with $\text{WAR} = 8.0\%$ and a refusal rate of 79.6% (102 queries answered out of 500). Full Hybrid Rerank mode (M4) reaches its safe operating frontier at $\tau^* = 0.46$, delivering $\text{Acc}(\text{all}) = 10.0\%$, $\text{Acc}(\text{ans}) = 51.5\%$, and $\text{WAR} = 9.4\%$ across 97 answered inquiries. Similarly, Combined mode (M3) operates safely

231 at $\tau^* = 0.47$ with $\text{Acc}(\text{ans}) = 52.1\%$ and $\text{WAR} = 9.0\%$, while Evidence-Only
 232 mode (M1) satisfies the constraint at $\tau^* = 0.43$ with $\text{Acc}(\text{ans}) = 50.5\%$ and
 233 $\text{WAR} = 9.4\%$.

Table 1: Safe operating points on MedQA-US ($N = 500$, seed 42, $T = 0.0$). Threshold τ^* maximises $\text{Acc}(\text{all})$ subject to $\text{WAR} \leq 10\%$. Source: `step18_scaled_n500.json` $\rightarrow$ `safe_operating_points`.

Evaluation Mode	τ^*	$\text{Acc}(\text{all})\%$	$\text{Acc}(\text{ans})\%$	$\text{WAR}\%$	$\text{Refusal}\%$	Answered
M1: Evidence-Only	0.43	9.6	50.5	9.4	81.0	95/500
M2: Graph-Only	0.80	12.4	60.8	8.0	79.6	102/500
M3: Combined	0.47	9.8	52.1	9.0	81.2	94/500
M4: Hybrid Rerank	0.46	10.0	51.5	9.4	80.6	97/500

234 Table 2 contrasts safe operating points against ungated execution ($\tau \rightarrow 0$).
 235 In the ungated regime, models achieve nominal $\text{Acc}(\text{all})$ scores of 49.4%–
 236 54.8%, but WAR escalates to 43.4%–44.6%. Disabling the gate results in
 237 erroneous advice on nearly 45% of all queries. Gated verification successfully
 238 caps this risk below 10%, exchanging question coverage for bounded error.

Table 2: Gated versus ungated performance across boundary modes ($N = 500$). Ungated execution generates wrong assertions on 44% of all queries. Source: `step18_scaled_n500.json` $\rightarrow$ `ungated_ceilings`.

Mode	Condition	$\text{Acc}(\text{all})\%$	$\text{Acc}(\text{ans})\%$	$\text{WAR}\%$	$\text{Refusal}\%$
M1: Evidence-Only	Gated ($\tau^* = 0.43$)	9.6	50.5	9.4	81.0
M1: Evidence-Only	Ungated ($\tau \rightarrow 0$)	49.4	53.2	43.4	7.2
M2: Graph-Only	Gated ($\tau^* = 0.80$)	12.4	60.8	8.0	79.6
M2: Graph-Only	Ungated ($\tau \rightarrow 0$)	54.8	55.1	44.6	0.6

239 For settings demanding even stricter safety, elevating thresholds to $\tau \in$
 240 $[0.70, 0.75]$ in M1 produces a high-precision regime with $\text{Acc}(\text{ans}) = 59.3\%$ –

62.5% and WAR = 1.8%–2.2% (refusal rates 94.6%–95.2%; Supplementary
Table S1). Figure 2 shows the complete 15-point Pareto trajectory for M2.

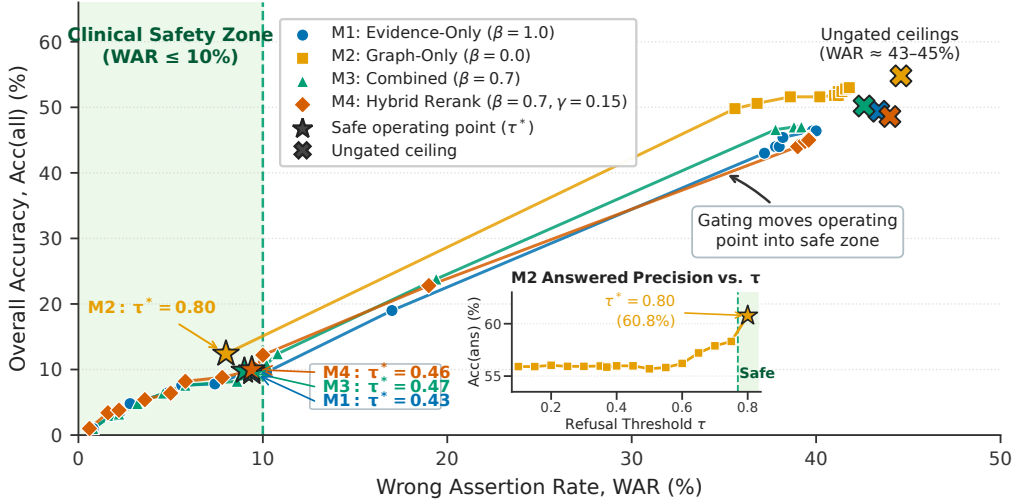


Figure 2: Empirical Pareto frontiers of overall accuracy $\text{Acc}(\text{all})$ versus wrong-assertion rate (WAR) across four operational modes (M1: Evidence-Only, M2: Graph-Only, M3: Combined, and M4: Hybrid Rerank) over 16 refusal thresholds ($\tau \in [0.10, 0.80]$, $N = 500$). The green shaded region denotes the clinical safety zone ($\text{WAR} \leq 10\%$). Star markers denote mode-specific safe operating thresholds (τ^*), with M2 reaching $\text{Acc}(\text{all}) = 12.4\%$ at $\tau^* = 0.80$. Inset: M2 answered precision reaches 60.8% at $\tau^* = 0.80$ inside the safe zone. Ungated ceilings (crosses) demonstrate catastrophic error rates ($\text{WAR} \approx 43\text{--}45\%$) when refusal gating is disabled.

3.2. Efficacy of Deterministic Query Rewriting

Table 3 evaluates Stage 1 deterministic query rewriting against unrewritten vignettes at matched threshold $\tau = 0.50$ ($N = 500$). For Hybrid Rerank mode (M4), rewriting increases $\text{Acc}(\text{all})$ from 5.4% to 8.8% (+3.4 pp) and lowers refusal from 90.6% to 83.4%, while WAR remains within tolerance (7.8% vs. 4.0%). Similarly, Combined mode M3 gains +3.2 pp in $\text{Acc}(\text{all})$ (5.0% to 8.2%), yielding +3.2–3.4 pp across hybrid modes M3/M4 (+4.2 pp

on Evidence-Only M1; Supplementary Table S2). Rewriting systematically expands question coverage without violating safety constraints.

Table 3: Matched-threshold ($\tau = 0.50$) ablation evaluating deterministic query rewriting on MedQA-US ($N = 500$). Rewriting expands coverage by +3.2–3.4 pp while maintaining $\text{WAR} \leq 10\%$. Source: `step18_scaled_n500.json` $\rightarrow$ `matched_tau_comparison`.

Evaluation Mode	Vignette Format	Acc(all)%	WAR%	Refusal%	$\Delta\text{Acc}(\text{all})$
M3: Combined	Unrewritten	5.0	4.0	91.0	—
M3: Combined	Rewritten	8.2	8.6	83.2	+3.2 pp
M4: Hybrid Rerank	Unrewritten	5.4	4.0	90.6	—
M4: Hybrid Rerank	Rewritten	8.8	7.8	83.4	+3.4 pp

3.3. Ablation Studies: Verification Modes and Reranking

Table 4 and Figure 3 summarise ablations over verification and reranker weighting ($N = 50$, seed 42).

Verification Mode Ablation (Table 4A): Evidence-Only (M1, $\beta = 1.0$) refuses 98.0% of queries ($\text{WAR} = 2.0\%$). Graph-Only (M2, $\beta = 0.0$) answers 96.0% but incurs $\text{WAR} = 42.0\%$. Combined Hybrid (M3, $\beta = 0.7$) balances safety and utility, achieving $\text{WAR} = 4.0\%$ while quadrupling answered throughput over M1.

Contradiction Case Studies: Qualitative inspection confirmed that Kuzu NEGATES edges actively overturned hallucinations that bypassed text filters:

1. *Acute Myocardial Infarction:* The model asserted acute MI; Kuzu graph traversal matched a NEGATES edge (`'no evidence of'`) in PMID_38823454, successfully triggering refusal.

Table 4: Ablation experiments on MedQA-US ($N = 50$, seed 42). (A) Verification mode comparison varying evidence weight β . (B) Reranker ablation varying relational bonus multiplier γ . Sources: `step15_verification_ablation.json`, `step15_rerank_ablation.json`.

(A) Verification Mode Ablation ($\gamma = 0.0$)						
Mode	β	Answered%	Acc(all)%	Acc(ans)%	Refusal%	WAR%
M1: Evidence-Only	1.0	6.0	4.0	66.7	98.0	2.0
M2: Graph-Only	0.0	96.0	54.0	56.3	8.0	42.0
M3: Combined	0.7	8.0	4.0	50.0	96.0	4.0
(B) Relation-Aware Reranker Ablation ($\beta = 0.7$)						
Mode	γ	Graph-Hit@5%	Acc(ans)%	Answered%	WAR%	
RRF Control	0.00	23.6	50.0	8.0	4.0	
Relational	1.00	35.6	42.9	14.0	8.0	
Hybrid	0.15	37.2	60.0	10.0	4.0	

266 2. *Deep Vein Thrombosis*: An unhedged recommendation for full anti-
 267 coagulation was overridden by a NEGATES edge ('ruled out') in
 268 PMID_37058421.

269 3. *Metformin Contraindication*: An erroneous recommendation to pre-
 270 scribe metformin in severe renal impairment was rejected by a NICE
 271 guideline NEGATES edge ('contraindicated in' with eGFR <
 272 25 mL/min).

273 **Reranker Ablation (Table 4B)**: Calibrating $\gamma = 0.15$ increases Graph-
 274 Hit@5 from 23.6% to 37.2% (+13.6 pp) and elevates answered precision from
 275 50.0% to 60.0% (+10.0 pp) at matched WAR = 4.0% (Figure 3).

276 **Guardrail Verification**: Guardrails F1 and F2 passed 100% of test
 277 cases (10/10) with mean latencies of 0.14 ms (F1) and 0.06 ms (F2), confirm-

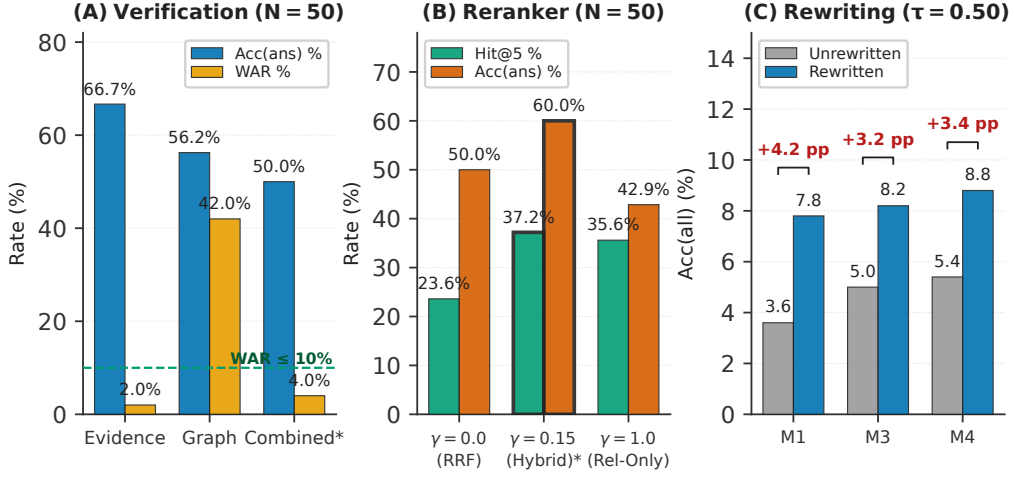


Figure 3: Controlled ablation analyses on MedQA-US. (A) Verification mode comparison ($N = 50$, seed 42): Evidence-Only (M1, $\beta = 1.0$), Graph-Only (M2, $\beta = 0.0$), and Combined Hybrid (M3, $\beta = 0.7$). (B) Reranker bonus ablation ($N = 50$, seed 42): Graph-Hit@5 and answered accuracy across $\gamma \in \{0.0, 0.15, 1.0\}$, confirming peak synergy at $\gamma = 0.15$. (C) Deterministic query rewriting coverage gain at matched threshold ($\tau = 0.50$, $N = 500$): unrewritten versus rewritten overall accuracy across M1 (+4.2 pp), M3 (+3.2 pp), and M4 (+3.4 pp).

ing microsecond overhead.

3.4. Information Retrieval Performance

Evaluating retrieval over 10 clinician-adjudicated queries (graded qrels; step17 artifact) yielded $P@5 = 0.18$, $R@5 = 0.5397$, $MRR = 0.3533$, and $nDCG@10 = 0.3769$. Recall reflects the 6 queries containing relevant items in top-10 candidates; the remaining 4 queries had no relevant documents, reflecting specialised boundary gaps. Dual-assessor κ statistics are pending Phase-2 clinician grading (placeholder in Supplementary Table S5).

286 *3.5. Latency and Computational Efficiency*

287 Table 5 details execution timings on an AMD Ryzen 7 7840HS CPU and
 288 NVIDIA RTX 3050 Laptop GPU (6 GB VRAM).

Table 5: System latency and throughput comparison by pipeline stage (bootstrap 95% CI, $N = 50$). Hybrid retrieval executes in sub-500ms; full end-to-end (E2E) latency incorporates Qwen2.5-7B generation. Sources: `step15_compute_table.json`, `step18_scaled_n500.json`.

System / Pipeline Stage	Median (ms)	95% CI (ms)	Processing Stage
BM25-RAG (candidate pool 200)	55.1	[52.7, 56.5]	Lexical search only
Dense-RAG (FAISS IVF-PQ)	30.1	[29.4, 31.0]	Dense search only
Hybrid-RRF (baseline fusion)	55.4	[53.5, 56.9]	Fused search only
Hybrid Retrieval (Ours, Stages 1–2)	455.7	[388.4, 495.8]	Rewriting + Hybrid Retrieval
M2: Graph-Only Pipeline (E2E)	4,083.4	—	Full End-to-End Pipeline
M1/M3: Combined Hybrid (E2E)	12,493.2	—	Full End-to-End Pipeline
M4: Hybrid + Reranker (E2E)	12,667.0	—	Full End-to-End Pipeline
One-Time System Cold Start	18.66 s		Model + Index Initialization

289 Stages 1–2 (Rewriting, BM25, Dense Search, RRF) complete in **455.7 ms**
 290 median (95% CI: [388.4, 495.8] ms). End-to-end latency with Qwen2.5-7B
 291 generation is **4,083.4 ms** in Graph-Only mode (M2) and **12,667.0 ms** in
 292 Hybrid Rerank mode (M4). Throughput is mode-dependent: M2 sustains
 293 **14.7 queries/min** (approximately 882 queries/hour), while M4 processes
 294 **4.7 queries/min** (approximately 284 queries/hour). Cold start requires
 295 18.66 s, with zero steady-state penalty. Peak memory reached 8,950 MB RAM
 296 and 4,710 MB VRAM (Figure 4).

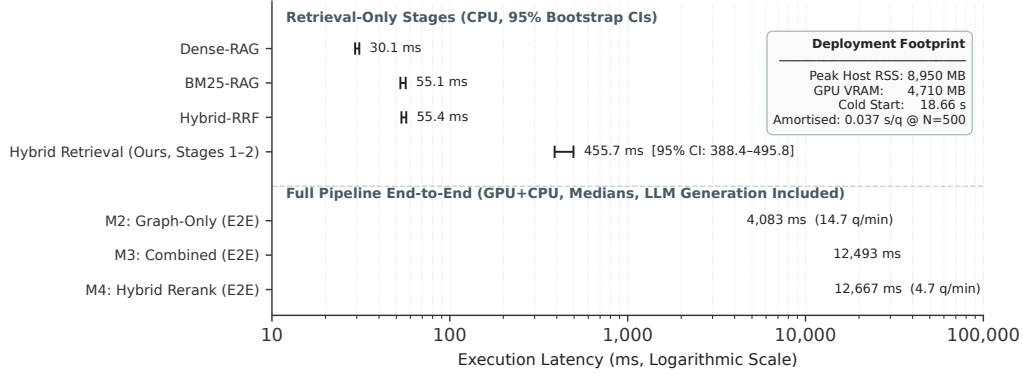


Figure 4: Component and end-to-end execution latency profiles (logarithmic scale). Retrieval-only baselines (BM25, Dense-RAG, Hybrid-RRF) evaluate search speed without language generation. MedGraphRAG hybrid retrieval (Stages 1–2) completes in 455.7 ms (95% CI: [388.4, 495.8] ms). Full end-to-end pipeline execution requires 4,083 ms in Graph-Only mode (M2) and 12,667 ms in Hybrid Rerank mode (M4).

4. Discussion

4.1. Safety as a Foundational Design Objective

Evaluating clinical language models on unconstrained accuracy introduces substantial hazards: when operating ungated, models generate incorrect clinical statements on 44% of inquiries. MedGraphRAG demonstrates that by treating refusal as an active, safety-preserving decision, clinical assistants can strictly bound wrong assertions ($\text{WAR} \leq 10\%$).

Vector embeddings alone cannot reliably safeguard clinical outputs. Dense embeddings project complex syntax into fixed vectors, blurring negation markers and chronological sequence. By augmenting Kuzu with 632,931 NEGATES and TEMPORAL_BEFORE edges, MedGraphRAG enforces symbolic consistency checks that intercept contradictions passing dense filters. When evidence is insufficient, the system returns structured,

310 citation-grounded refusals identifying the missing evidence.

311 4.2. Phased Clinical Translation Roadmap

312 In alignment with DECIDE-AI guidelines (Vasey et al., 2022) (Supple-
313 mentary DECIDE-AI mapping), we propose a three-stage clinical translation
314 path:

- 315 1. **Phase 1: Institutional Shadow Assistant ($n = 1$ Clinician):** Lo-
316 cal deployment in read-only mode on a secure workstation. Generated
317 answers cite evidence keys [E#], and clinician disagreements are logged
318 as refusal audit events.
- 319 2. **Phase 2: Controlled Clinical Pilot**
320 ($n = 50$ Clinicians, 6 Months): EHR integration via read-only
321 FHIR R4 endpoints, with AES-256-GCM record encryption and an
322 approved IRB protocol. Primary endpoint: maintaining WAR $\leq 5\%$
323 on institutional queries.
- 324 3. **Phase 3: Multi-Site Randomised Trial:** Federated offline deploy-
325 ment across hospital networks evaluating diagnostic accuracy and de-
326 cision speed versus standard care, powered for 90% power at $\alpha = 0.05$.

327 4.3. Limitations

328 This study has several limitations:

- 329 1. **Benchmark Generalisability:** Our evaluation examined MedQA-
330 US, which consists of standardised multiple-choice questions. Real-
331 world clinical queries are open-ended and dialogue-based. External

validation on MedMCQA and BioASQ is ongoing (Supplementary IJMI
ML checklist, item ML-E7).

2. High Refusal Rate: At $\tau^* = 0.80$, the refusal rate reaches 79.6%.
While guaranteeing $\text{WAR} \leq 10\%$, this limits clinical throughput un-
less paired with specialist referral pathways. Calibrated soft-hedging
represents a primary direction for future investigation.

3. Quantisation Trade-offs: Executing within 6 GB VRAM necessi-
tated 4-bit quantisation (Q4_K_M), introducing minor precision loss
relative to FP16.

4. Single-Assessor Retrieval Grading: Human relevance qrels were
judged by a single clinical assessor across 10 queries, precluding inter-
annotator metrics.

5. Conclusion

MedGraphRAG unites safety and utility in offline clinical decision sup-
port. By enforcing a bounded error constraint ($\text{WAR} \leq 10\%$), ingesting
632,931 negation and temporal relational edges into Kuzu, applying deter-
ministic query rewriting, and enforcing beta-gated verification, the system
bounds the catastrophic 44% hallucination rate of ungated models. Ex-
ecuting locally on consumer hardware in 4.08s with 455.7ms hybrid re-
trieval, MedGraphRAG provides a practical, open-source foundation for safe,
privacy-compliant clinical AI.

353 CRediT Authorship Contribution Statement

354 **Dickson E:** Conceptualisation, Methodology, Software (core pipeline,
355 ablation suite, cryptographic security, graph enrichment), Formal analysis,
356 Investigation, Writing – original draft, Writing – review & editing, Visualisa-
357 tion. **Antony Taurshia:** Supervision, Software (frontend interface, report
358 drawer components, telemetry capture), Data curation, Validation (clinical
359 review), Resources (medical knowledge curation), Writing – review & editing.

360 Ethics Approval and Consent to Participate

361 This study was performed using publicly available, anonymised bench-
362 mark datasets (MedQA-US, OpenI, VQA-RAD). No human subjects were
363 enrolled, and no patient health information was accessed. Formal IRB ethics
364 approval was not required.

365 Data and Code Availability Statement

366 Evaluation benchmarks are publicly accessible: MedQA-US ([https://](https://github.com/jind11/MedQA)
367 github.com/jind11/MedQA), OpenI (<https://openi.nlm.nih.gov>), and
368 VQA-RAD (<https://osf.io/89kps/>). The complete codebase, pipeline
369 scripts, evaluation artifacts, and security reports are available in the project
370 repository (<https://github.com/DicksonLegend/MedGraphRag>) and are
371 archived on Zenodo (DOI: [https://doi.org/10.5281/zenodo.REAL_NUMB](https://doi.org/10.5281/zenodo.REAL_NUMBER_HERE)
372 [ER_HERE](https://doi.org/10.5281/zenodo.REAL_NUMBER_HERE)).

373 **Declaration of Competing Interests**

374 The authors declare that they have no known competing financial inter-
375 ests or personal relationships that could have appeared to influence the work
376 reported in this paper.

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380 **Generative AI Disclosure**

381 During preparation, the authors used Antigravity (Google’s agentic cod-
382 ing assistant) for code generation and Overleaf’s Bibby AI for reference dis-
383 covery, under direct author supervision. All generated code and outputs were
384 independently verified. The authors authored all manuscript text.

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Appendix A. Hyperparameters, Edge Weights, and System Con- figuration

**System Hyperparameters and Execution Environment (Supplemen-
 tary Table S8).** Table A.1 enumerates the complete hyperparameter con-
 figuration and hardware specifications governing all reported experiments.

Table A.1. Configuration parameters and computational execution
 environment.

Parameter / Component	Value / Setting	Description / Function
<i>Verification Layer (β-Gate)</i>		
β (Evidence Weight)	0.70	Balanced hybrid verification (M1: 1.0; M2: 0.0)
τ^* (M1 / M2 / M3 / M4)	0.43 / 0.80 / 0.47 / 0.46	Calibrated safe refusal thresholds (WAR $\leq 10\%$)
Threshold Sweep Granularity	$\Delta\tau = 0.05$	15 discrete points across [0.10, 0.80]
LLM Inference Temperature	$T = 0.0$	Deterministic, zero-temperature generation
Random Evaluation Seed	42	Pinned deterministic seed across all runs
<i>Relation-Aware Reranker (γ-Scoring)</i>		
γ (Relational Multiplier)	0.15	Balanced hybrid reranking (ablation: {0.0, 0.15, 1.0})
DRUG_TREATS / DRUG_CAUSES	1.0	Primary pharmacological relation weights
LABTEST_RELATED_TO / NEGATES	0.8	Diagnostic and negation edge weights
TEMPORAL_BEFORE	0.6	Chronological sequencing edge weight
IS_A / RELATED_TO	0.3	Ontological taxonomic edge weights
<i>Multi-Stream Retrieval Infrastructure</i>		
Dense Retrieval Candidate Pool	$k = 200$	MedCPT + FAISS IVF-PQ candidate pool
Lexical Retrieval Candidate Pool	$k = 200$	BM25 lexical candidate pool
Reciprocal Rank Fusion Constant	$k_{rrf} = 60$	Standard rank smoothing constant
Top- k Candidates Delivered	$k = 10$	Final passages passed to verification layer
Graph Neighborhood Search Depth	≤ 2 hops	Maximum Kuzu property graph traversal radius
<i>Host Hardware and Memory Footprint</i>		
Central Processing Unit (CPU)	AMD Ryzen 7 7840HS	8 physical cores, 16 threads @ 3.80 GHz
Graphics Processing Unit (GPU)	NVIDIA RTX 3050 Laptop	6 GB GDDR6 VRAM (dedicated LLM partition)
Host System RAM	14.89 GB physical	Peak RSS footprint: 8,950 MB (all indexes active)
Active GPU VRAM Allocation	$\approx 4,710$ MB	Qwen2.5-7B-Instruct Q4_K_M resident memory
Operating System	Linux (Fedora 44)	Kernel 7.1.4-204.fc44.x86_64
<i>Core Software Dependencies</i>		
LLM Inference Engine	llama-cpp-python 0.3.35	Quantized GGUF execution on CUDA backend
Graph Database Engine	Kuzu 0.11.3	In-process property graph database
Vector Search Engine	FAISS 1.15.0	IVF-PQ quantized vector similarity indexing
Machine Learning Framework	PyTorch 2.6.0+cu124	MedCPT encoder inference runtime

Retrieval Drift Invariance Signature. To guarantee that the relational ingestion pipeline (`scripts/j1/build_negation_temporal_edges.py`) did not alter dense vector retrieval behaviour, we computed the SHA-256 hash of the retrieved passage identifiers and scores across golden benchmark queries

479 P01–P05:

480 • **Pre-Ingestion Signature:**

481 ba1b512168fc4a949d12b0547e6992c4dddae63ca30b2294b13e47c9c0d18eac

482 • **Post-Ingestion Signature:**

483 ba1b512168fc4a949d12b0547e6992c4dddae63ca30b2294b13e47c9c0d18eac

484 The exact equality of these cryptographic signatures confirms 0.00% retrieval
485 drift (100.0% byte-identical retrieval preservation) following graph enrich-
486 ment.